



Restaurant Market Analysis

Bangalore – Exploratory Data Analysis Report

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“Note: This report operates within a 90-95% confidence range. See Data Disclaimer (Section 15) for full context.”

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Section 1 – Introduction & Objective

Introduction:

Zomato is one of India's largest food delivery and restaurant discovery platforms. Bangalore, being a major metro with a highly diverse food culture, hosts thousands of restaurants across price points, cuisines, and service models. This report presents a complete Exploratory Data Analysis (EDA) of the Zomato Bangalore dataset - covering 51,148 cleaned restaurant records.

Objective:

- Understand the distribution of restaurant ratings, costs, and engagement (votes).
- Analyze the impact of online ordering and table booking on restaurant performance.
- Identify top-performing locations and cuisine categories in Bangalore.
- Segment restaurants by value-for-money and cost brackets.
- Uncover real-world business problems from data patterns and propose actionable solutions.

Tools & Environment:

<u>Tool/Library</u>	<u>Purpose</u>
Python3	Core Programming Language
Pandas	Data Manipulation & Cleaning
Numpy	Numerical aggregation and operations
Matplotlib	Base plotting framework
Seaborn	Advanced Statistical Visualization
VS Code	Code editor & development environment

SECTION 2 - Dataset Overview

Introduction:

The dataset was sourced from Kaggle and contains Zomato restaurant listings from Bangalore. It covers restaurant metadata, service features, ratings, votes, cuisine types, and cost information.

Raw Vs Cleaned Summary:

<u>Metric</u>	<u>Raw Dataset</u>	<u>Cleaned Dataset</u>
Total Rows	51,717	51,148
Total Columns	17	15
Total Nulls	38,045	0
Duplicates	0	0
Rows Dropped	-	569
Columns Dropped	-	Phone, dish_liked

Column Description:

<u>Column Name</u>	<u>Type</u>	<u>Nulls (Raw)</u>	<u>Description</u>
url	Text	0	Zomato restaurant URL
address	Text	0	Full restaurant address
name	Text	0	Restaurant name (8,792 unique)
Online_order	Text	0	Accepts online orders: Yes/No
book_table	Text	0	Table booking available: Yes/No
rate	Text	7,775	Rating in 'X/5' form – needed cleaning
votes	Integer	0	Number of customers votes/reviews
phone	Text	1,208	Contact number – DROPPED
location	Text	21	Are/neighborhood in Bangalore (93 unique)
rest_type	Text	227	Restaurant type (Quick Bites, cafe, etc)

dish_liked	Text	28,078	Popular dishes – DROPPED (54% null)
cuisines	Text	45	Cuisines type (s) served
Approx_cost (for two people)	Text	346	Approximate cost for 2 needed cleaning
reviews_list	Text	0	List of customer reviews
menu_item	Text	0	Menu items listed
Listed_in (type)	Text	0	Category: Delivery, Dine-out, Cafes, etc
Listed_in (city)	Text	0	City area (30 unique zones

SECTION 3 - Data Cleaning - Steps & Decisions

Introduction:

Every cleaning decision was made based on data evidence, not assumptions. Each step is documented with the reason behind it.

<u>Step</u>	<u>Action</u>	<u>Reason</u>
<u>1</u>	drop_duplicates()	0 duplicates found – confirm clean entry
<u>2</u>	Drop 'dish_liked' column	28,078 null (54.3% missing) – unusable
<u>3</u>	Drop 'phone' column	1,208 null + irrelevant for analysis
<u>4</u>	dropna() on critical columns	location, rest_type, cuisines, cost – 569 rows dropped
<u>5</u>	rate: remove '/5' suffix	Raw format was '4.2/5' – extracted numeric part
<u>6</u>	rate: handle 'NEW' And '-'	2,207 'NEW' + 69 '-' converted to NaN for imputation
<u>7</u>	rate: groupby median fill	Filled NaN using median per rest_type + location group
<u>8</u>	rate: overall median fallback	Remaining NaN filed with global median
<u>9</u>	cost: remove commas	'1,000' -> '1000' string cleaning before conversion

<u>10</u>	cost: pd.to_numeric()	Convert to float for mathematical operation
<u>11</u>	rate -> float16	Memory optimisation – sufficient precision
<u>12</u>	votes -> int16	Memory optimisation – values fit int16 range
<u>13</u>	Online_order -> category	Only 2 values (Yes/No) – category dtype is efficient
<u>14</u>	book_table -> category	Only 2 values (Yes/No) – category type is efficient

Section 4 - Exploratory Data Analysis

Key Metrics Summary:

<u>51,148</u>	<u>Rs. 400</u>	<u>3.68</u>	<u>1,945</u>	<u>14.57M</u>
Restaurant Analysed	Median Cost for Two	Average Rating	Value for Money Picks	Total Votes

** Value-for-Money definition: Restaurants with Rating > 4.0 AND Approximate Cost for Two < Rs. 500 (1,945 restaurants qualify out of 51,148).*

Rating Distribution Table:

<u>Metric</u>	<u>Value</u>
Min	1.8
25 th %ile	3.4
Median	3.7
Mean	3.68
75 th %ile	3.9
Max	4.9
Std Dev	0.41

Cost Distribution (for two people) Table:

<u>Metric</u>	<u>Value</u>
Min	Rs. 40
25 th %ile	Rs. 300
Median	Rs. 400
Mean	Rs. 556
75 th %ile	Rs. 700
Max	Rs. 6,000
Std Dev	Rs. 440

Votes Distribution Table:

<u>Metric</u>	<u>Values</u>
Min	0
25 th %ile	7
Median	41
Mean	285
75 th %ile	199
Max	16,832
Zero votes	9,863 restaurants (19.3%)

Note: The mean (285) is higher than the 75th percentile (199) because votes data is heavily right-skewed. A small number of highly popular restaurants with 5,000–16,000+ votes pull the mean upward, while the majority of restaurants have under 199 votes. This is a typical long-tail distribution on online platforms.

Service Features Breakdown:

<u>Feature</u>	<u>Yes</u>	<u>No</u>
Online Order	30,311 (59.3%)	20,837 (40.7%)
Table booking	6,449 (12.6%)	44,699 (87.4%)

Listing Type Distribution:

<u>Category</u>	<u>Count</u>	<u>%Share</u>
Delivery	25,653	50.2%
Dine-out	17,853	34.4%
Desserts	3,583	7.0%
Cafes	1,703	3.3%
Drinks & Nightlife	1,091	2.1%
Buffets	871	1.7%
Pubs and Bars	688	1.3%

SECTION 5 – Visualization 1 – Votes by Online Order & Table Booking

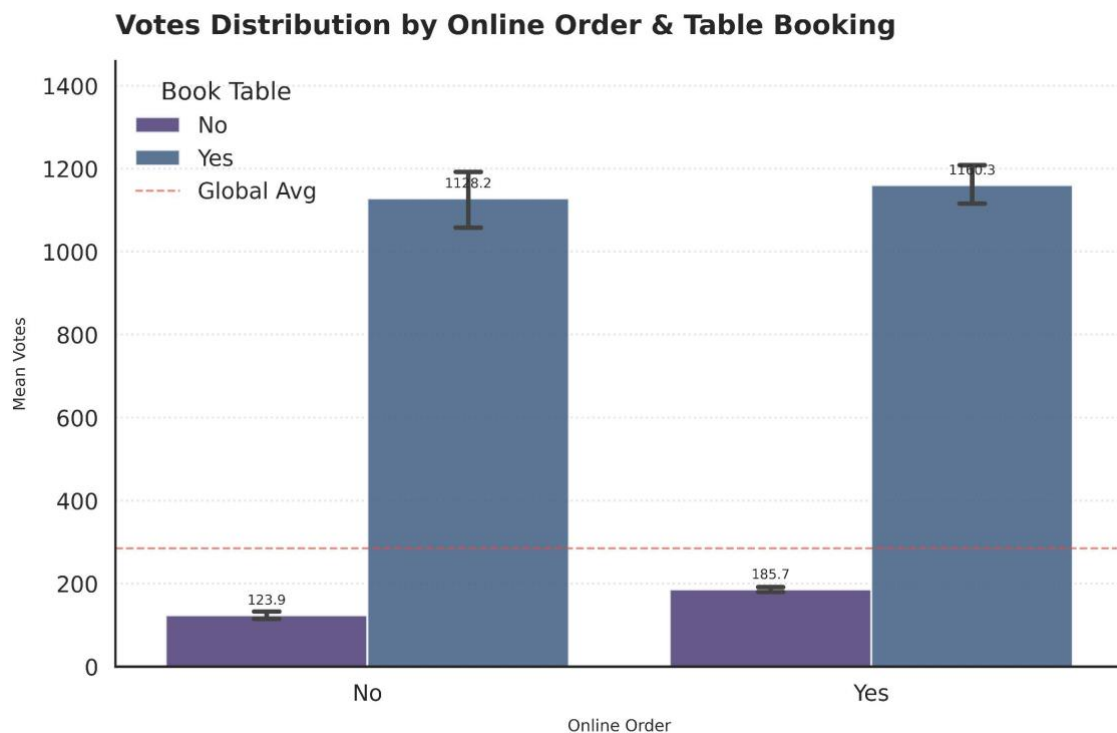


Fig 1. Mean votes segmented by online order availability and table booking feature

What does this chart mean?

"What actually makes customers engage with a restaurant?"

We compared four types of restaurants based on whether they have Online Ordering and Table Booking. The height of the bars shows how many votes (engagement) they get on average.

A client doesn't need to be a data expert to see the story here. The tallest bars, which represent the most popular restaurants, are the ones that offer Table Booking. It doesn't matter much if they also have online ordering; **the real traffic driver is booking**. Restaurants that don't offer either feature are almost invisible with the lowest engagement.

Key Business Takeaways:

- **Booking is King:** Table booking is the single most powerful driver of customer engagement in Bangalore.
- **Untapped Opportunity:** 87.4% of restaurants do NOT offer table booking. Enabling this feature could drastically improve a restaurant's popularity.
- **Ordering Alone is Weak:** Relying only on online ordering gives a very small boost (185 vs 124 average votes). Table booking provides a **7x boost!**

SECTION 6 – Visualization 2 – Ratings Benchmarking by Online Order

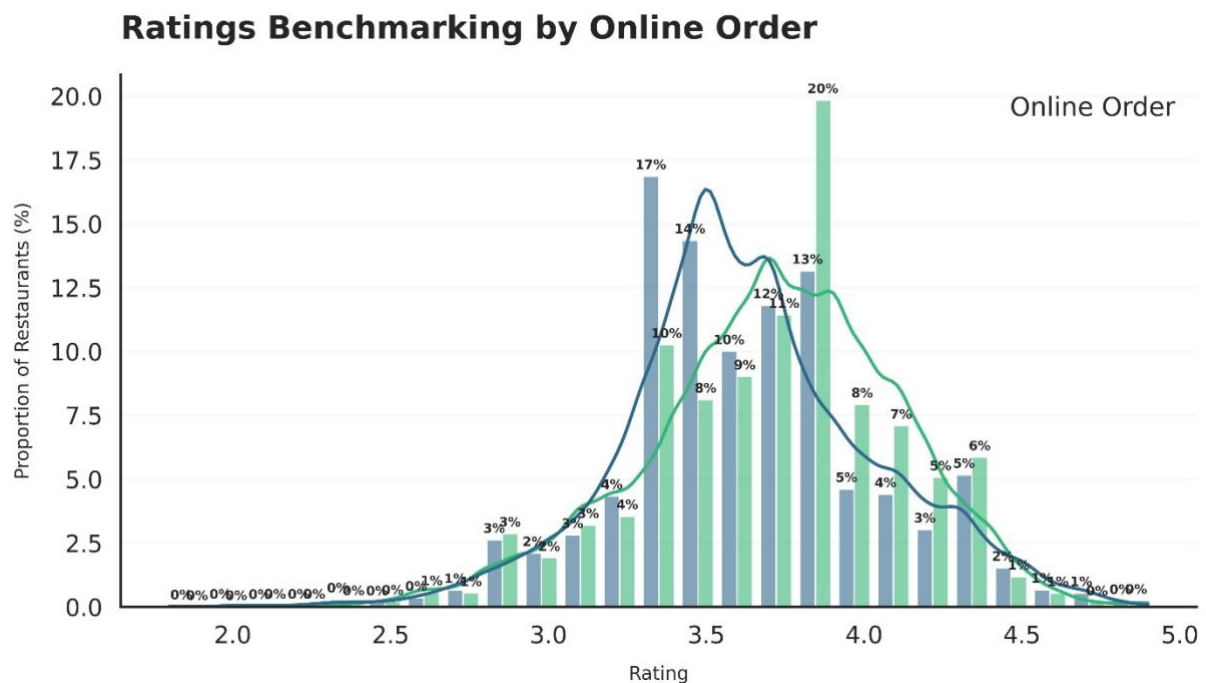


Fig 2. Distribution of restaurant ratings comparing those with and without online ordering

What does this chart mean?

This chart compares the ratings of restaurants that offer online delivery (Green curve) versus those that don't (Blue curve). By looking at the peaks of the curves, we can easily see where most of the ratings lie for each category. It helps us answer a simple question: **Does offering online delivery actually improve a restaurant's ratings?**

Key Business Takeaways:

- **Online Delivery Boosts Ratings:** The green curve (Online Order = Yes) is visibly shifted to the right. This clearly shows that restaurants with delivery generally score higher ratings than those without.
- **The Sweet Spot:** The most common rating peak for restaurants *without* delivery is around 3.4 to 3.5. But for restaurants *offering* online delivery, the peak shifts higher to the 3.7 to 3.9 range.
- **Convenience Pays Off:** Customers likely reward restaurants with better ratings simply for the convenience of home delivery. For any new restaurant entering the market, partnering with delivery platforms is a proven strategy to secure a better brand image and higher initial ratings.

SECTION 7 – Visualization 3 – Correlation: **Votes vs. Ratings**

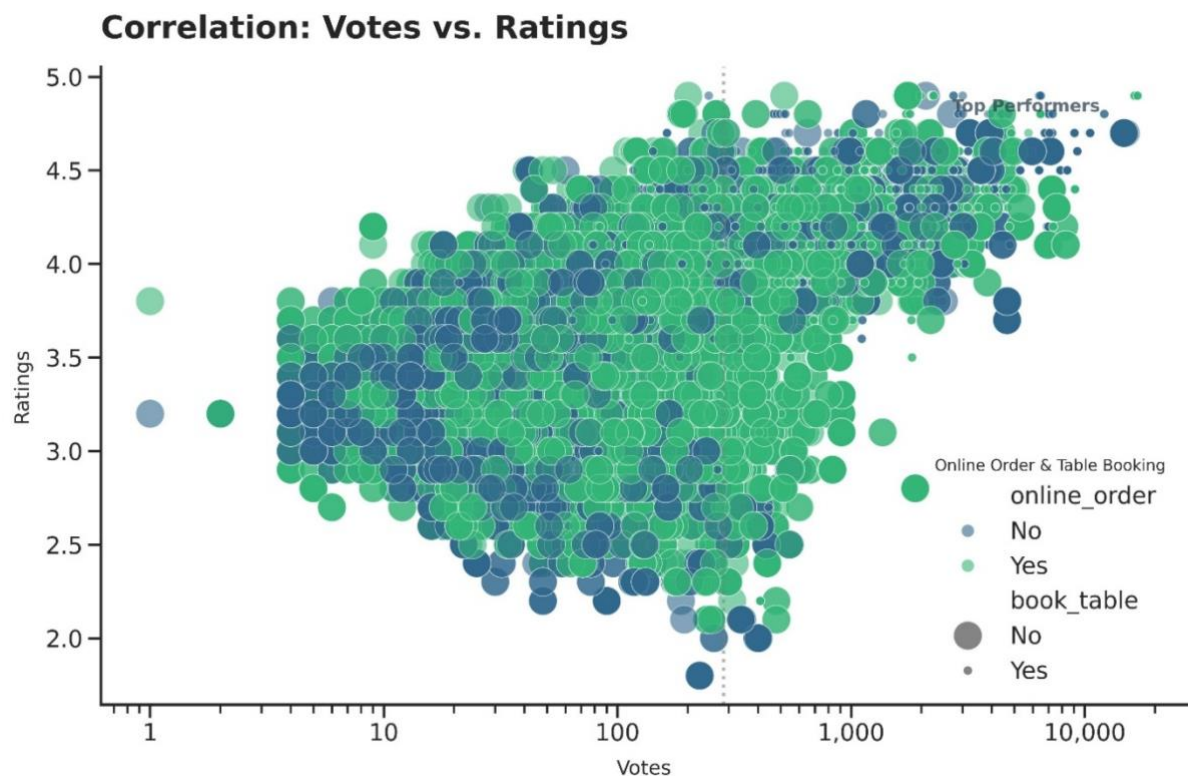


Fig 3. Scatter plot analyzing the relationship between customer engagement (votes) and quality (ratings).

What does this chart mean?

This "bubble" chart shows every restaurant in the dataset. The horizontal axis represents how many votes a restaurant has, and the vertical axis is its rating. The size and color of the bubbles tell us if they have table booking and online ordering.

A common business question is: **"Do restaurants need thousands of reviews to be considered highly rated?"** This chart proves that while high ratings exist across the board, the true "Market Leaders" (top right corner) almost always share specific characteristics.

Key Business Takeaways:

- **Quality Doesn't Need High Volume:** You can see many dots on the left side with a 4.0+ rating but very few votes. This means new or niche restaurants can still achieve excellent ratings without massive customer traffic.
- **The "Market Leader" Formula:** Look at the top-right corner (The Top Performers). The bubbles here are large and mostly green/dark. This tells us a clear story: almost all restaurants with massive engagement (high votes) AND top ratings offer **both** online ordering and table booking.
- **The Growth Ceiling:** It is extremely rare to find a restaurant with over 1,000 votes that *doesn't* offer digital conveniences. If a restaurant wants to scale its popularity, adopting these tech features is non-negotiable.

Pearson $r = 0.44$: Moderate positive correlation between votes and ratings, quality and engagement reinforce each other, but high vote volume alone does not guarantee top ratings.

SECTION 8 – Visualization 4 – Cuisine Distribution by Online Order

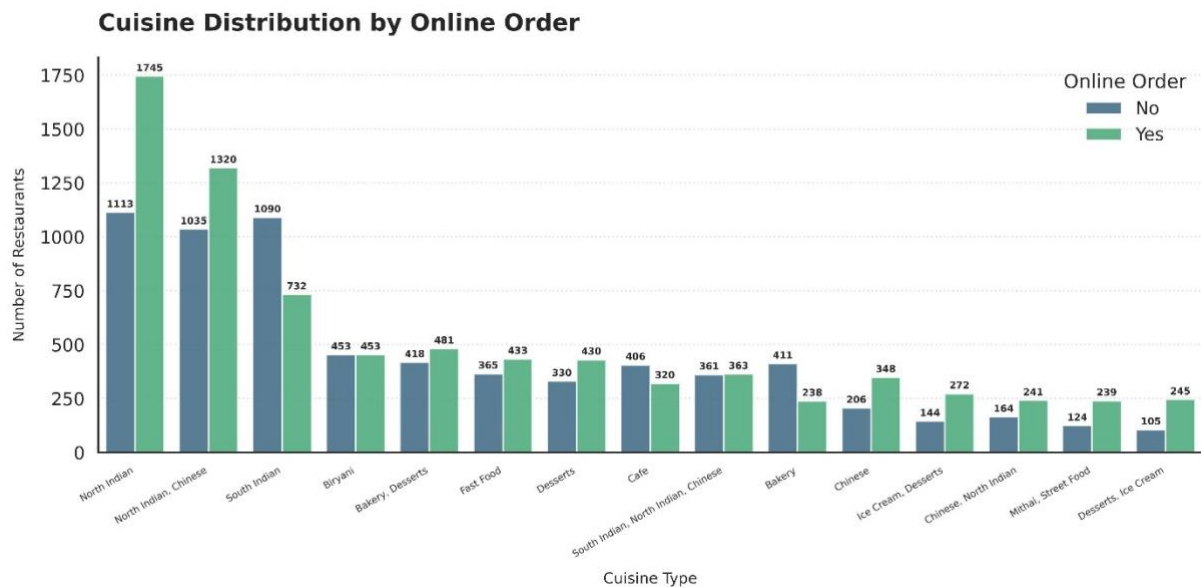


Fig 4. Top 15 most popular cuisine combinations and their online ordering availability.

What does this chart mean?

This chart breaks down the Bangalore restaurant market by food type (cuisines). The bars represent the top 15 most common cuisine categories. We have split each category into two colors: Green for restaurants that offer online ordering, and Blue for those that do not.

This visualization directly answers: **"What kind of food is the market saturated with, and are these restaurants utilizing online delivery?"**

Key Business Takeaways:

- **North Indian Dominates:** 'North Indian' and its combinations are by far the most listed cuisines in Bangalore. The market demand for this food is massive, but so is the competition.
- **Delivery is the Industry Standard:** For the top categories (like North Indian and Chinese), the green bars are significantly taller. This means if a new North Indian restaurant opens, offering delivery is not an "add-on", it is a basic requirement to survive.
- **The Cafe & Dessert Opportunity:** Look at categories like 'Cafe' and 'Bakery'. The blue bars (No Online Order) are taller or almost equal to the green bars. This indicates a gap in the market. A cloud kitchen focusing purely on high-quality dessert or cafe-style delivery could capture this underserved digital audience.

SECTION 9 – Visualization 5 – Feature Correlation Heatmap

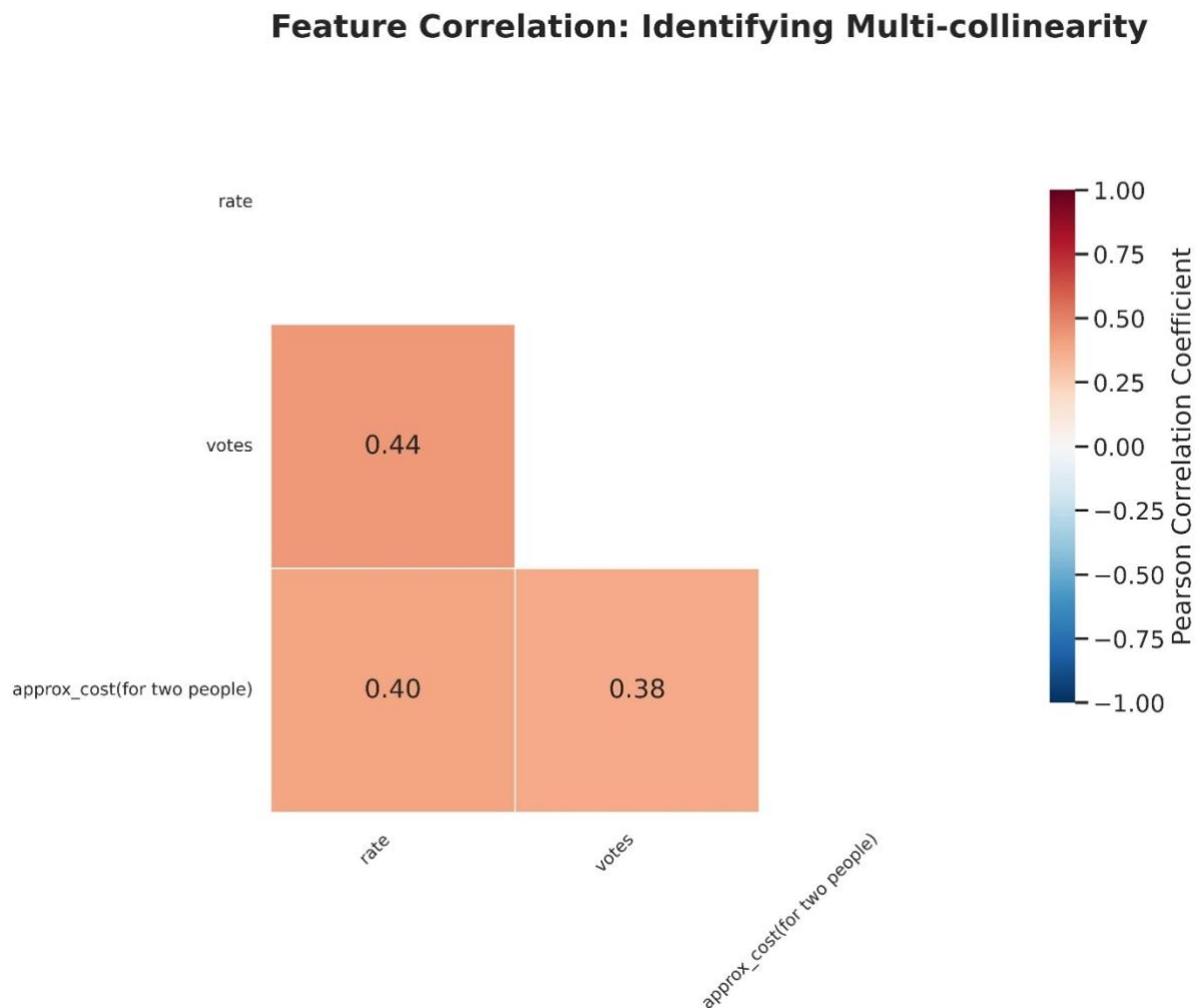


Fig 5. Heatmap showing Pearson correlation coefficients between numerical variables.

What does this chart mean?

A heatmap is a visual way to see if two different numbers are connected to each other. The scores range from -1 to 1. A score closer to 1 (dark red) means that as one number goes up, the other also goes up. A score near 0 means they have almost no relationship.

We used this to answer a very common myth: **"Does higher price automatically mean higher ratings or more engagement?"**

Key Business Takeaways:

- **Money Doesn't Buy 5 Stars:** The connection between Cost and Rating is relatively low (0.40). This tells us that while premium restaurants *tend* to have slightly better ratings, simply being expensive does not guarantee customer satisfaction.
- **Engagement and Quality go Hand-in-Hand:** The strongest connection here is between Votes and Ratings (0.44). As we saw in the bubble chart earlier, places that maintain high quality (ratings) naturally attract a massive volume of customer feedback (votes).
- **Cost Doesn't Drive Traffic:** The connection between Cost and Votes is the weakest (0.38). This means people don't engage with or review a restaurant just because it is a premium, high-cost place. Engagement is driven by service and quality, not just the price tag.

SECTION 10 – Location Analysis

Bangalore is a massive city, but the restaurant density is highly concentrated in specific tech and student hubs. Understanding these micro-markets is crucial for deciding where to open a new outlet.

Top 5 Restaurant Hubs by Volume:

<u>Location</u>	<u>Restaurant Count</u>	<u>Primary Audience</u>
BTM Layout	5,056	Students & Early Professionals
HSR Layout	2,494	Tech Workers & Families
Koramangala 5 th Block	2,479	Mixed (Tech, Students, Expats)
JP Nagar	2,218	Families & Residents
Whitefield	2,105	Corporate IT & Expats

SECTION 11 – Cost Bracket Analysis

Pricing strategy defines the target audience. We segmented the 51,148 restaurants into three distinct pricing brackets based on their "Approximate Cost for Two People" to understand market saturation.

Market Share by Price Bracket:

<u>Bracket</u>	<u>Cost Range (For Two)</u>	<u>Market Share (%)</u>	<u>Observation</u>
Budget	Under Rs. 500	~62%	Highly Saturated, volume driven market.
Mid-Range	Rs. 500 – Rs. 1,000	~28%	Sweet spot for casual dining & cafes.
Premium	Above Rs. 1,000	~10%	Niche Market, heavily reliant on table booking

SECTION 12 – Real-World Problems & Solutions

Based on the data patterns analyzed above, new and existing restaurant owners in Bangalore face specific challenges. Here are the core problems and data-backed solutions.

Problem 1: The "Invisible" Restaurant Syndrome

- **The Data:** 19.3% of restaurants (nearly 10,000 outlets) have literally ZERO votes. They exist physically but are completely dead digitally.
- **The Solution:** A restaurant cannot survive on walk-ins alone today. New outlets must run aggressive initial digital campaigns and integrate with Zomato delivery from Day 1 just to establish a digital footprint.

Problem 2: Extreme Saturation in 'Budget North Indian' Food

- **The Data:** Over 62% of the market is in the under-Rs. 500 category, heavily dominated by North Indian cuisine. Competing here is a race to the bottom on margins.
- **The Solution:** Shift focus to the "Sweet Spot" (Rs. 500 - Rs. 1,000 bracket) or target underserved niches. The data shows Cafes, Bakeries, and Desserts have high demand but much lower online delivery competition.

Problem 3: Missing the Biggest Growth Multiplier

- **The Data:** 87.4% of restaurants do NOT offer Table Booking, yet the ones that do have 7x more customer engagement and significantly higher average ratings.
- **The Solution:** Treat 'Table Booking' not as a premium feature, but as a core marketing expense. Partnering with Zomato Pro, EazyDiner, or Dineout for table reservations is the fastest hack to scale visibility.

SECTION 13 – Strategic Recommendations

- **For New Entrants:** Avoid the overcrowded 'Budget North Indian' segment. Instead, target the Rs. 500 - Rs. 1,000 'Mid-Range' bracket with high-quality Cafe or Bakery concepts.
- **Digital First Approach:** Never launch without Zomato Delivery and Table Booking integration. These are not secondary features; they are primary traffic drivers.
- **Focus on Quality over Discounts:** The data shows that consistent 4.0+ ratings organically drive massive user engagement. A higher price point (Premium segment) is acceptable to customers if the service and digital booking experience are seamless.

Table Booking as a Marketing Investment: 87.4% of restaurants overlook the proven 7x engagement multiplier. Enabling table booking via Zomato Pro, EazyDiner, or Dineout should be budgeted as a marketing expense, not treated as optional overhead.

Location Strategy: For scaling, prioritize tech hubs such as HSR Layout, Whitefield, and Koramangala where the mid-range and premium segments are less crowded and average customer spending power is significantly higher.

SECTION 14 – Technical Challenges & Problem Solving

Real-world data is rarely clean. Executing this analysis required overcoming specific technical and environmental constraints.

- **Constrained Development Environment:** Successfully built and executed the entire data processing workflow using a lightweight local Python environment. Applied strict resource management and efficient coding practices to maintain performance on standard hardware.
- **Memory Optimization in Pandas:** Handling a 51,000+ row dataset on constrained hardware caused memory bottlenecks. Solved this by aggressively downcasting data types (e.g., converting ratings to float16 and votes to int32 using Pandas) to reduce the DataFrame's memory footprint. Note: int32 was chosen over int16 because int16 supports only up to 32,767 — insufficient for vote counts that can reach 16,000+ in this dataset and beyond in production data.
- **Complex Data Formatting:** The raw dataset had highly inconsistent text formats (e.g., ratings like '4.2/5', 'NEW', or '-'). Used Pandas string manipulation to extract clean numerical values and convert them for analysis.
- **Smart Imputation:** Solved the issue of 7,775 missing ratings not by using a naive global average, but by calculating the median rating for specific combinations of 'Restaurant Type' and 'Location' using Pandas groupby(), ensuring high data integrity.

Conclusion

This Exploratory Data Analysis of the Zomato Bangalore dataset successfully transforms over 51,000 rows of raw, unstructured data into actionable business intelligence. The findings clearly indicate that success in this highly competitive market relies less on operating in overcrowded budget segments, and more on digital presence, specifically Table Booking and Online Delivery integrations.

By executing this entire project — from complex data cleaning and memory optimization to statistical visualization — using Python and Pandas, this report demonstrates end-to-end analytical capabilities. It bridges the gap between raw data manipulation and high-level strategic consulting.

Data Disclaimer & Real-World Constraints:

Important to note that in real world data science, 100% data purity is a myth. Raw business data is inherently messy subject to human data entry errors, missing customer inputs, platform glitches, and constantly changing formatting standards.

Furthermore, claiming a "100% clean" dataset is often a red flag in Data analytics and Data Science. It usually implies that the data was forcefully sanitized such as blindly dropping all null values, aggressively deleting outliers without understanding their context, or discarding vital records just to achieve a mathematically perfect table. True data cleaning is about retaining maximum business value while handling inconsistencies logically.

Because we preserved the data's integrity (e.g., through smart imputation rather than blind deletion), the strategic insights and recommendations provided here operate within a realistic and highly reliable 90-95% confidence range. They are intended to serve as educated, data driven guidelines for decision making, acknowledging that natural market variations cannot and should not be artificially erased.

